

METEHAN GÜNEN

Network Administrator

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🌐 Metrohan 🔗 LinkedIn 🌐 Website



Summary

Computer Engineering graduate specializing in backend development, Linux systems, IoT, computer vision, and real-time applications. Experienced in designing and deploying end-to-end systems that integrate scalable backend services, embedded hardware, cloud infrastructure, and AI-based decision pipelines. Built production-ready platforms using technologies including FastAPI, Node.js, React, PostgreSQL, MongoDB, Docker, and Raspberry Pi. Strong background in system architecture, performance optimization, network administration, technical documentation, and solving real-world engineering problems across healthcare, research, and event technology domains.

Professional Experience

Network Administrator, *Pievision*

04/2026 – Present
İstanbul, Türkiye

- Managing network infrastructure and ensuring system reliability and uptime
- Monitoring system performance and troubleshooting network-level issues
- Supporting deployment and maintenance of backend services and internal tools

Researcher, *i-Lab*

08/2025 – Present
İstanbul, Türkiye

- Developing AI-powered biomedical and IoT systems with real-time processing constraints
- Designed end-to-end pipelines combining embedded devices, cloud services, and computer vision
- Worked on segmentation and detection systems using YOLO and edge computing approaches
- Collaborated in multidisciplinary teams to prototype and validate research-driven solutions

Intern, *BGTS*

07/2025 – 08/2025
İstanbul, Türkiye

- Built scalable backend services using Node.js & Express with asynchronous architecture
- Designed REST APIs and improved system performance and maintainability
- Integrated PostgreSQL and implemented structured data handling
- Contributed to system architecture decisions and participated in code reviews

Education

B.Sc. Computer Engineering, *Istinye University*

08/2022 – 06/2026
İstanbul, Türkiye

Technical Skills

Programming

Python, C, C++, SQL, JavaScript

IoT & Embedded Systems

Raspberry Pi, ESP32, Arduino, Sensor Integration

Tools & Platforms

Docker, Git/GitHub, Linux, Nginx, Prometheus, Grafana

Backend & Databases

Node.js, Express, FastAPI, REST APIs, PostgreSQL, MongoDB

Computer Vision/AI

YOLO, PyTorch, TensorFlow, OpenCV, scikit-learn

Languages

Turkish

Native/Bilingual

English

Proficient

Projects

CERAHIS, *Cerrahi Alet Hasar Tespit ve Raporlama Sistemi*

- Capstone project with Semih Çelenk, advised by Dr. Fevzi Aytaç Durmaz (Istinye University / i-Lab)
- Designed an end-to-end system for automated surgical instrument damage detection using YOLOv11m (mAP@50: ~0.99)

trained on 9,361 images across 13 instrument classes

- Built Raspberry Pi 3B+ + Arducam IMX519 hardware cabinet with a CrowPanel ESP32-S3 7" HMI display; optimized USB Serial

streaming from 0.2 FPS to ~15 FPS real-time preview

- Implemented FastAPI backend with confidence-based decision engine (auto_labeled / needs_review / unsure) and React

dashboard with inventory management and reporting modules

- Integrated QR/Datamatrix barcode reading via image processing for instrument identification
- Conducted FMEA risk analysis; system architecture designed for ISO 13485 and IEC 62304 compliance

Patent application filed (pending) — Türk Patent ve Marka Kurum

MEDHOLO

- TÜBİTAK 2209-B with Semih Çelenk industry-supported research project developed at i-Lab in collaboration with Pievision Ar-Ge

- AI-powered hospital triage kiosk using a 3D holographic avatar (Three.js) for contactless, hygienic patient interaction at

emergency entrances

- Implemented a hybrid decision architecture: deterministic reflex module for life-threatening keywords (< 0.01s response) + Local

LLM (Qwen2.5) for complex symptom reasoning

- Built Python/GPU-accelerated VAD → STT → LLM → TTS pipeline with lip-sync avatar rendering; "self-healing" infrastructure

resilient to network outages

- Achieved 100% functional end-to-end prototype (v25.0); future roadmap includes E-Nabız integration, biometric pain-scale

analysis, and multilingual support

TechEventRadar (eventradar.dev), *Turkish Tech Event Aggregator*

- Developed an open-source platform that centralizes Turkish technology conferences, meetups, workshops, competitions, and hackathons in a single searchable interface
- Published the project as open source to support community contributions and improve access to technology events across Türkiye